Exp:- 1

Program A :-

Manual calculation :-

9C + 7A (Addition)

1001 1100 (9C)
(+) 0111 1010 (7A)
1 0001 0110 = 16, carry = 1

(Subtraction) (9C - 7A)

2's complement of 7A

7A = 1000 0101 + 1
= 1000 0110 (86)

1001 1100 (9C)
(+) 1000 0110 (86)
1 0010 0010 = 22

carry out 1 → CY = 0 (no borrow)

Program :- B :-

Addition :- 1234 + 5678

Lower byte: 34 + 78 = AC (no carry)

Upper byte : 12 + 56 = 68

Sum = 68AC H, carry = 0

Subtraction: 1234 − 5678

L.B = 34 − 78 → 34 < 78, borrow 1

134 − 78 = BC

Upper Byte = 12 − 56 − 1 (borrow) → borrow again

112 − 56 − 1 = BB

Difference = BBBC H, Borrow = 1

5678 + BBBC = 11234

Program C:-

Multiplication by repeated addition 12H×05H

Count 5 → 0000 + 12 = 0012

Count 4 → 0012 + 12 = 0024

Count 3 → 0024 + 12 = 0036

Count 2 → 0036 + 12 = 0048

Count 1 → 0048 + 12 = 005A

Product = 005A H

In decimal 18×5 = 90 = 5AH

Division 64H ÷ 05H

64H = 100 (decimal) ; 05H = 5

05 is subtracted 20 times : 64 → 5F → 5A

5A → .... 05 → 00

Quotient = 20 = 14H, Remainder = 00H

14H × 05H + 00 = 64H

Program D:- 16-bit Multiplication & Division

Multiplication: 1234H × 5678H

1234H = 4660, 5678H = 22136

4660 × 22136 = 103,153,760 = 0626 0060H

U.W = 0626H

L.W = 0060H

Division (5678H ÷ 1234H)

5678 − 1234 = 4444 (Q=1)

4444 − 1234 = 3210 (Q=2)

3210 − 1234 = 1FDC (Q=3)

1FDC − 1234 = 0DA8 (Q=4)

0DA8 − 1234 = borrow, stop

Quotient = 0004H, Remainder = 0DA8H

Task 4:- (Division by Zero Check)

~~8-Bit:- 5A-127 and 5A-27~~ (No check)

HL = 1234, DE = 0000

Loop: 1234 − 0000 = 1234 → no borrow (CY=0)

1234 − 0000 = 1234 → no borrow (CY=0)

.... repeats forever

Zero check:-

D = 00, E = 00

A = D OR E = 00 OR 00 = 00 → Z=1

JZ Diver is taken → 4540 = FF

Valid case:- (1234H ÷ 0005H)

1234H = 4660 (decimal)

4660 ÷ 5 = 932 remainder 0

932 = 03A4H

Quotient = 03A4H, Remainder = 0000H

Experiment 2:-

Program A:-

· X = A7H , Y = 3CH

AND = 0010 0100 = 24H

OR = 1011 1111 = BFH

XOR = 1001 1011 = 9BH

x = 1010 0111 (A7)

y = 0011 1100 (3C)

The flag byte is 0001 0100 = 14H

8-bit complement:- A7H

x = 1010 0111 (A7)

NOTX = 0101 1000 = 58H (1's)

NOTX+1 = 0101 1001 = 59H (2's)

A7+59 = 1 00 → 00 with CY=1 //

16-bit complement W = 1234H:-

W = 0001 0010 0011 0100 (1234)

NOTW = 1110 1101 1100 1011, EDCBH

NOTW+1 = 1110 1101 1100 1100 = EDCCH

1234 + EDCC = 1 0000

· Program B:- Bit set/clear/Toggle/Test, Rotate & Shift

D = CAH = 11001010 (CBH)

Set bits 7,0: 1100 1010 OR 1000 0001 = 1100 1011

Clear bits 3:1: 1100 1010 AND 1111 0101 = 1100 0000 = C0H

Toggle low: 1100 1010 XOR 0000 1111 = 1100 0101 = C5H

Test bit 6: 1100 1010 AND 0100 0000 ≠ 0 → bit 6 = 1 → 01H

RLC : 1100 1010 → 1001 0101 = 95H

RRC : 1100 1010 → 0110 0101 = 65H

Logical left : 1001 0100 = 94H CY=1

Logical right : 0110 0101 = 65H CY=0

Arithemetic right : 1110 0101 = E5H

CA = -54 and -54 ÷ 2 = -27 = E5H

Count of 1s

1100 1010 → 1+1+0+0+1+0+1+0 = 4 → 04H

16 bit. Left (0AD H)

9234 = 1001 0010 0011 0100

X2 = 0010 0100 0110 1000 = 24684A

CY = 1 (D15 = 1 shifted out)